AKSHOBHYA MANTHATI

Austin, TX | Akshu.little@gmail.com | (737) 333-2997 | LinkedIn: akshobhya-manthati

EDUCATION

Westwood High School, Austin, TX 2022 – 2026
Austin Community College, Austin, TX 2023 – 2026

TECHNICAL SKILLS

- **Languages:** Python, Java, C++
- **Systems & Tools:** KiCad, NI Multisim, Git, Visual Studio Code, CAD (Fusion 360, Onshape, Inventor)
- **Fabrication & Prototyping:** CNC Machining, Laser Cutting, 3D Printing, Rapid Prototyping
- **Core Concepts:** PCB Design, Signal Integrity, PID Control, Hardware-Software Integration, DFM

ENGINEERING PROJECTS

Custom Mechanical Keyboard – PCB, Enclosure & Keycap Design

- Designing a custom keyboard PCB in KiCad, including switch matrix routing, controller integration, and production-ready board layout
- Developing the keyboard case and mechanical stack-up in CAD with attention to fit, mounting, manufacturability, and assembly.
- Conducted stack-up and mechanical tolerance analysis in CAD to ensure zero-interference component clearance and structural rigidity.

Piezoelectric Energy Harvesting Tile

- Designed KiCad PCB to harvest mechanical energy from footsteps, integrating ESP32 signal acquisition, custom filtering circuitry for improved SNR, and digital logging.
- 3D-printed load-bearing, hot-swappable tile enclosures to withstand foot traffic while uniformly distributing mechanical stress.
- Implemented battery charging, buck conversion, and aggressive firmware power-management states for optimal energy retention.

Python AI Voice Assistant – 1st Place Capstone (iStart Valley)

- Built a voice-controlled chatbot integrating OpenAI API with real-time speech recognition and response synthesis.

ROBOTICS EXPERIENCE

FTC - Cosmobotics Hardware Lead; Dean's List Semi-Finalist 2025

- Fabricated custom subassemblies using CNC-machined aluminum, laser-cut components, and 3D printing under strict competition deadlines.
- Designed drivetrain, intake, outtake, and claw subsystems; tuned PID control loops to maintain response stability under load.
- Developed a vision-based regression model to dynamically adjust shooter velocity, reaching Nationals Control Award and Top 10 International OPR.

WORK EXPERIENCE

Mathnasium Math Instructor

- Provided individualized math instruction across multiple grade levels, adapting lessons to student needs and building problem-solving confidence.

National Institute of Technology, Warangal Hybrid Energy Storage Research

- Collaborated with PhD researchers on hybrid energy storage systems for EVs and contributed to white paper market analysis.